

WHAT MAKES A HIT VIDEO GAME?



EXPLORING GLOBAL VIDEO GAME SALES TRENDS

Objective:

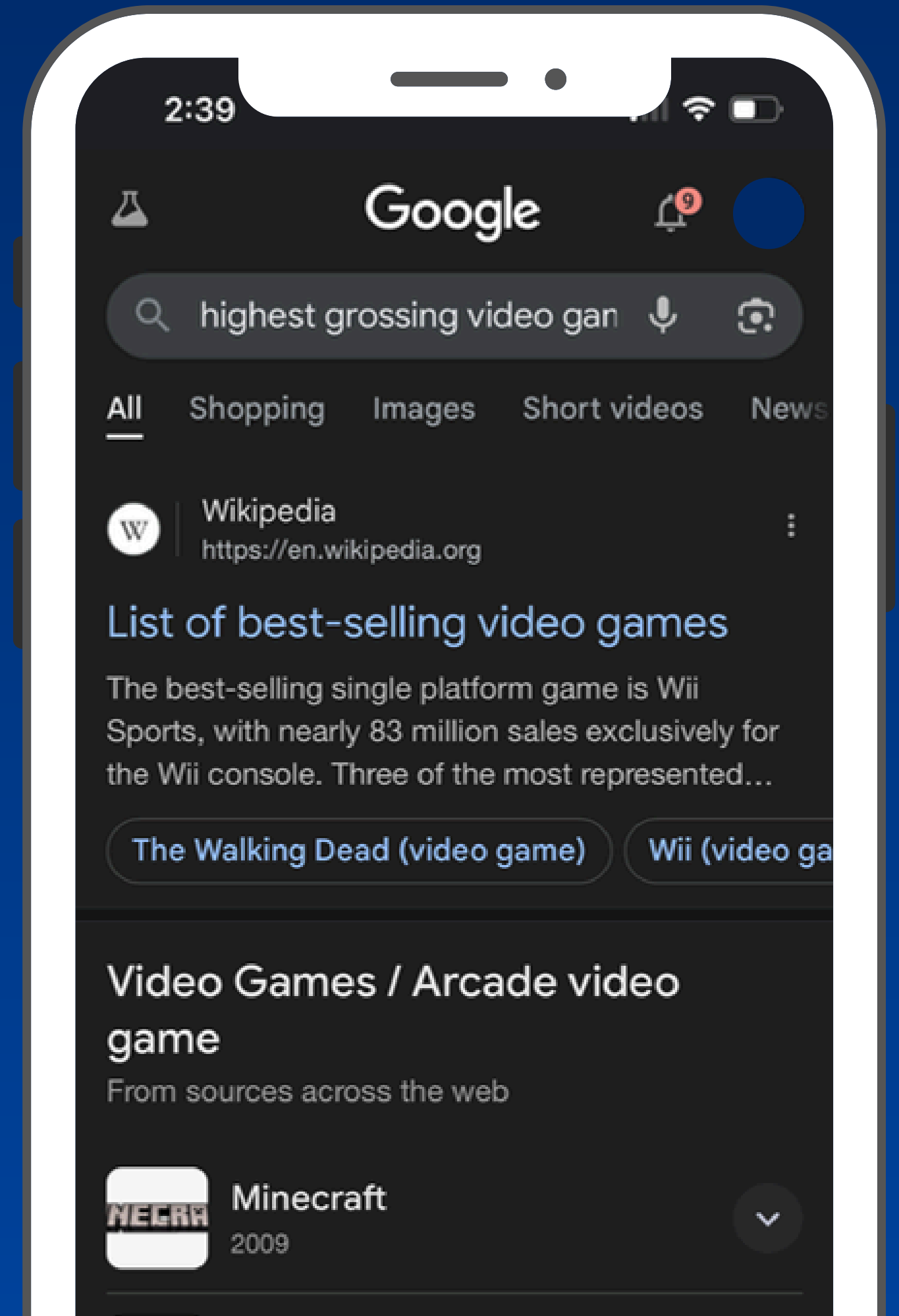
To analyze the sales data of global video games and determine key factors that contribute to a game's commercial success

Dataset Overview:

Includes information on game titles, platforms, genres, publishers, regional scores, critic scores, and release dates

Why this matters:

The video game industry is one of the most influential and fastest growing entertainment sectors. By uncovering patterns in sales performance, this project can help developers design games that align with player preferences, publishers make smarter investments based on trends, and analysts predict future hits using data driven models.



WHAT WE LOOKED FOR



HIGH RATINGS



POPULAR GENRE



HIGH SALES

We believe that these 3 parameters are the secret sauce to a games success, and with further analysis of our data, should be the most prominent aspects we see being expressed

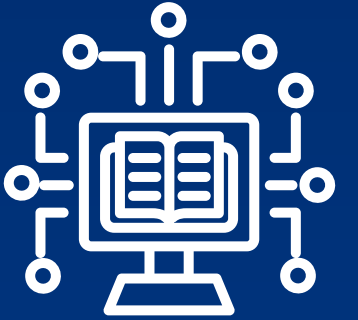
DATA EXPLORATION

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

Importing all
necessary
libraries for data
visualization

```
#drop the "developer, publisher, and title" features from the dataframe "df"
df = df.drop(['developer', 'publisher', 'title'], axis=1)
```

In order to work with the data effectively and present it in a more understandable way, we relied on various libraries designed for data manipulation and visualization. Tools like Pandas streamlined the process by offering a wide range of built-in functions that simplified the creation of visual outputs. By using Pandas alongside other supportive libraries, we were able to generate visual representations of the data collected from the CSV file, helping illustrate how the model could forecast certain results.



DATA CURATION

```
missing_data = df.isnull().sum()
missing_data = missing_data[missing_data > 0]
print("Columns with missing values:\n", missing_data)
```

✓ 0.0s

Columns with missing values:

developer	17
critic_score	57338
total_sales	45094
na_sales	51379
jp_sales	57290
pal_sales	51192
other_sales	48888
release_date	7051
dtype: int64	

```
df_cleaned = df.dropna()
```

```
print("Remaining missing values:", df_cleaned.isnull().sum().sum())
```

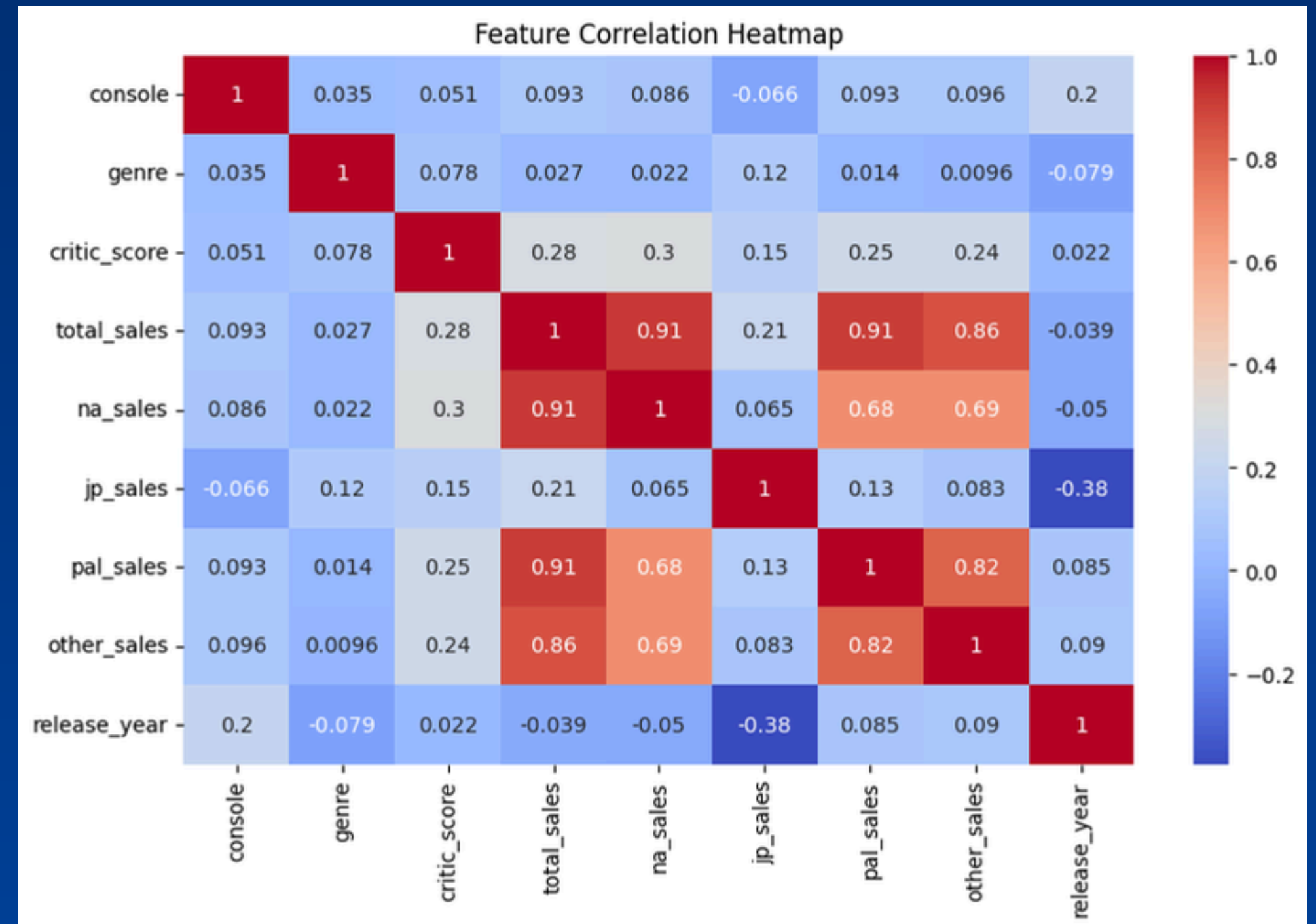
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Remaining missing values: 0

In order to ensure our data was as accurate as possible, we identified all the columns with missing values, and then cleaned them up so there would be no outliers when analyzing our data

DATA CURATION

As a way to point us in the right direction, we used a heatmap that included major features of the CSV file to help us identify the most closely related aspects. This visualization allowed us to quickly spot strong correlations and patterns within the data, guiding our focus toward the most impactful variables. From there, we were able to determine which features warranted deeper exploration and formed a solid foundation for creating meaningful and targeted illustrations.



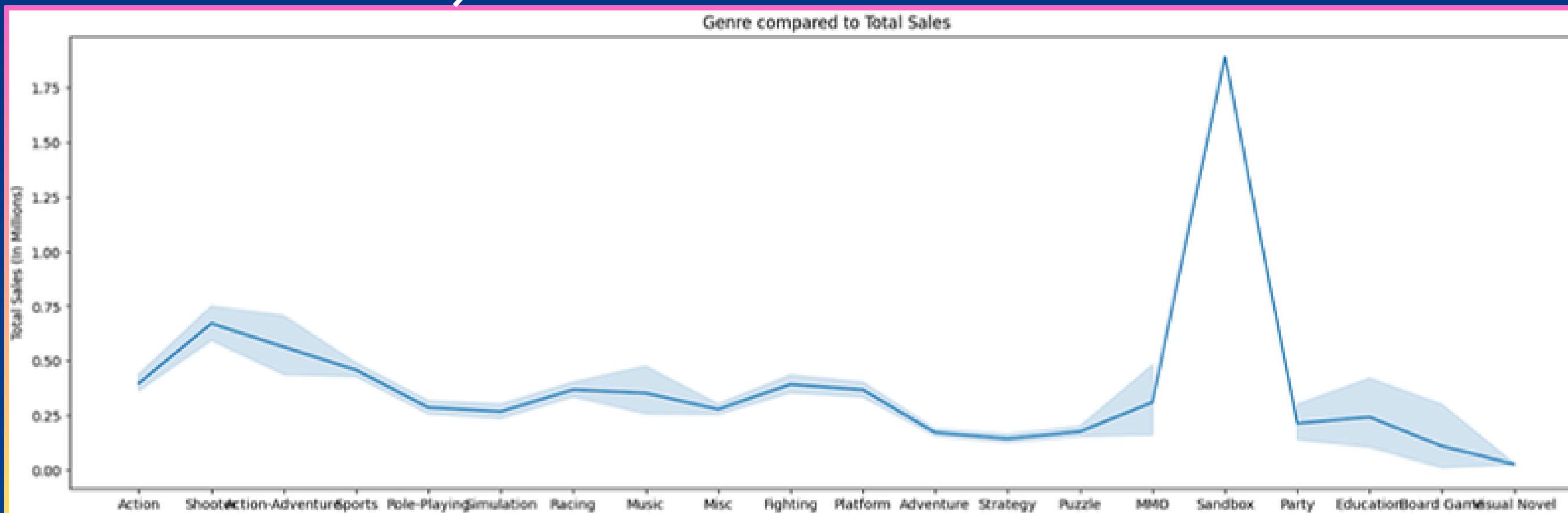
GENRE VS TOTAL SALES

```
[10] plt.figure(figsize = (20,6))
sns.lineplot(x = 'genre', y = 'total_sales', data = df)
plt.title("Genre compared to Total Sales")
plt.xlabel("Genre")
plt.ylabel("Total Sales (In Millions)")
```

VISUALIZATION

In order to determine how genre effects video game sales, we used the plt module in order to generate a line plot. As we can see from the line plot below, it seems that sandbox games generally tended to sell a lot more compared to other genres of video games.

The **Sandbox** genre had **1.75 million** sales, followed by **Shooters** with **750 thousand** sales, and **Fighting** with **500 thousand** sales respectively.

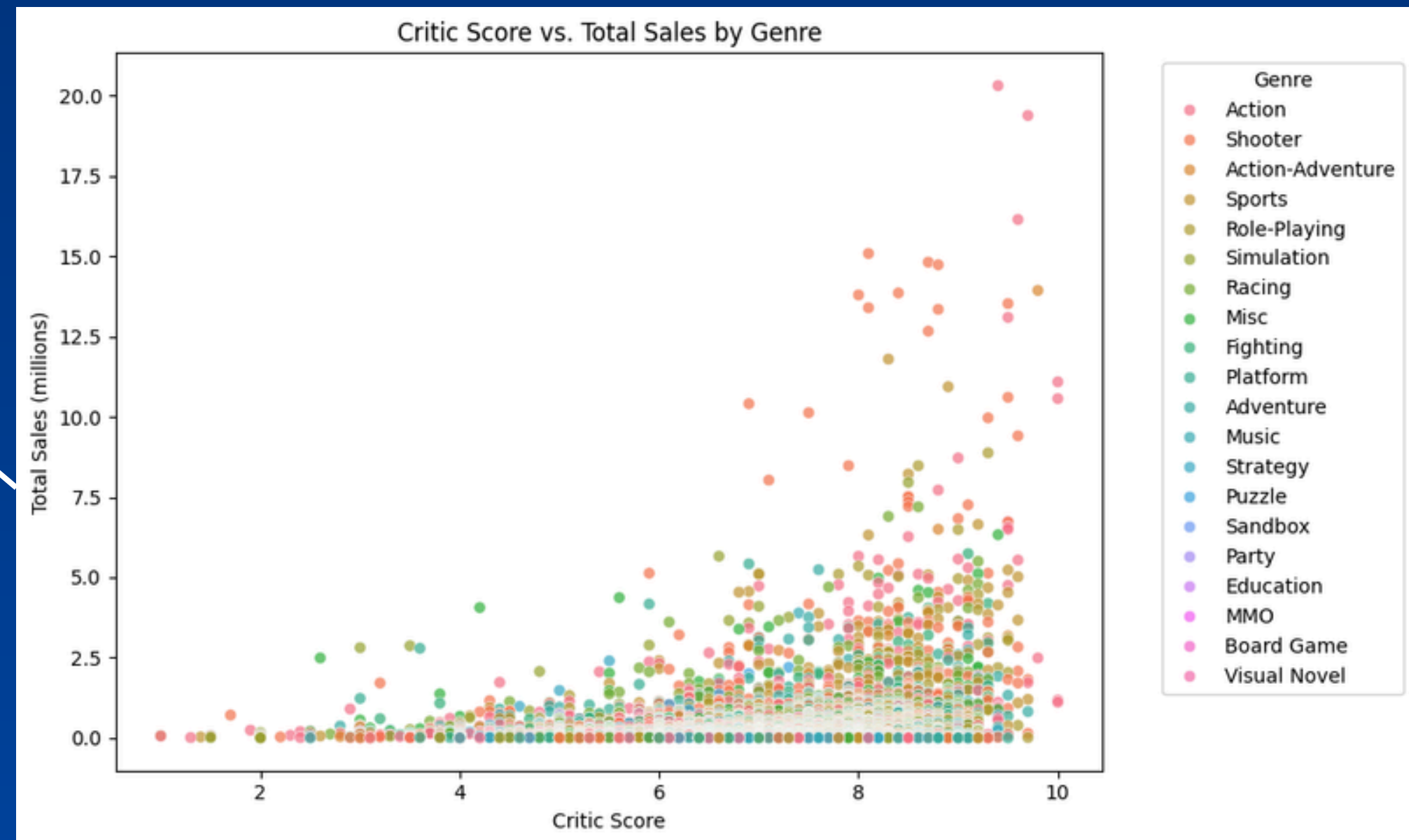


CRITIC SCORE AND TOTAL SALES

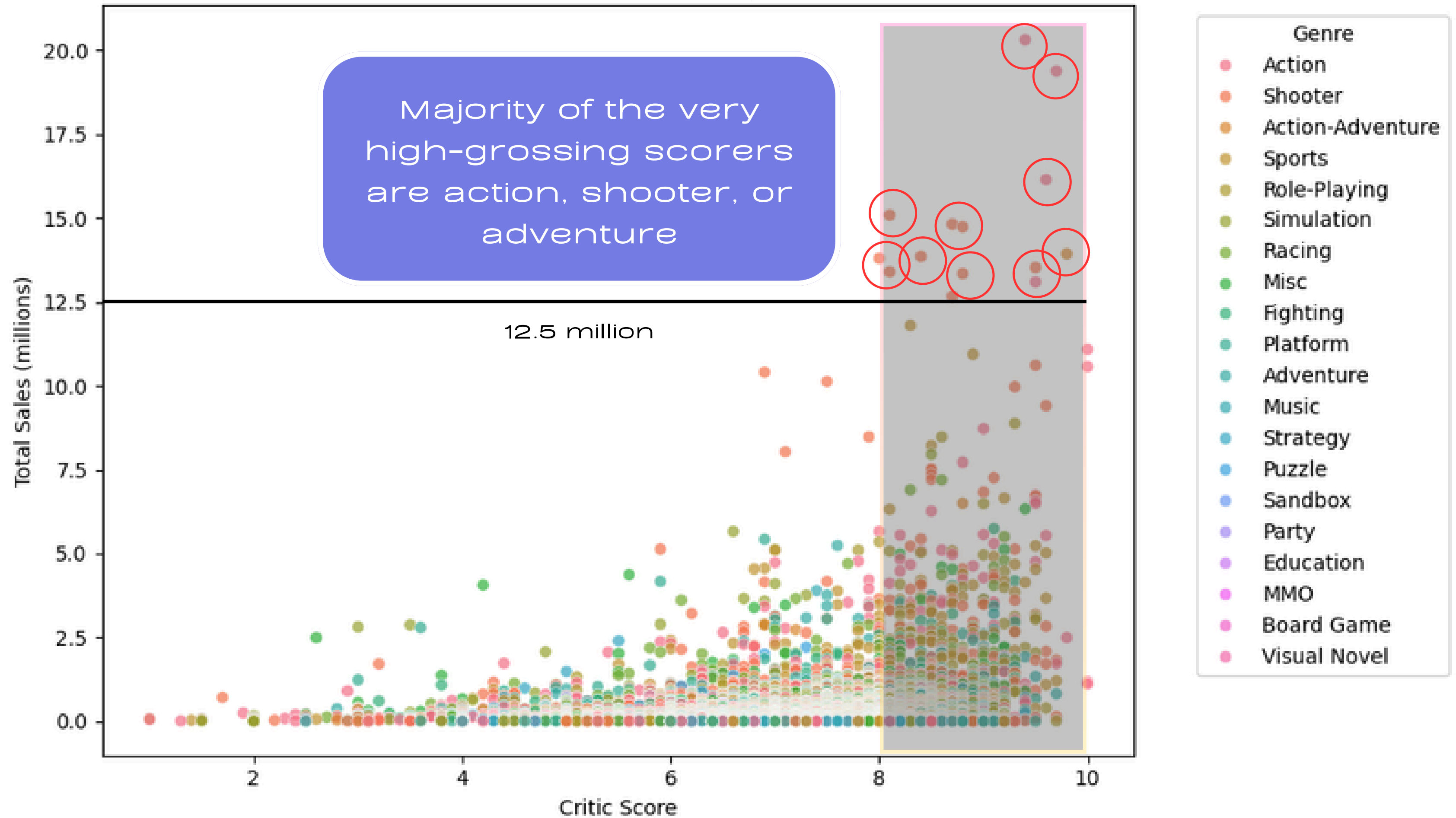
In order to see if the critic score had any correlation with the total sales, we then created a scatterplot to visualize the 2 data sets

As illustrated, games with scores between the range **8-10** had consistently **much higher total sales** compared to games with **lower scores**, so we can conclude that in order for a game to be **successful**, it should have a **high critic score**

```
plt.figure(figsize=(10, 6))
plt.scatter(scatter_df['critic_score'], scatter_df['total_sales'], alpha=0.5, s=20)
plt.title('Video Game Critic Scores by Total Sales')
plt.xlabel('Critic Score (0-10)')
plt.ylabel('Total Sales (Millions)')
plt.grid(True)
plt.show()
```



Critic Score vs. Total Sales by Genre

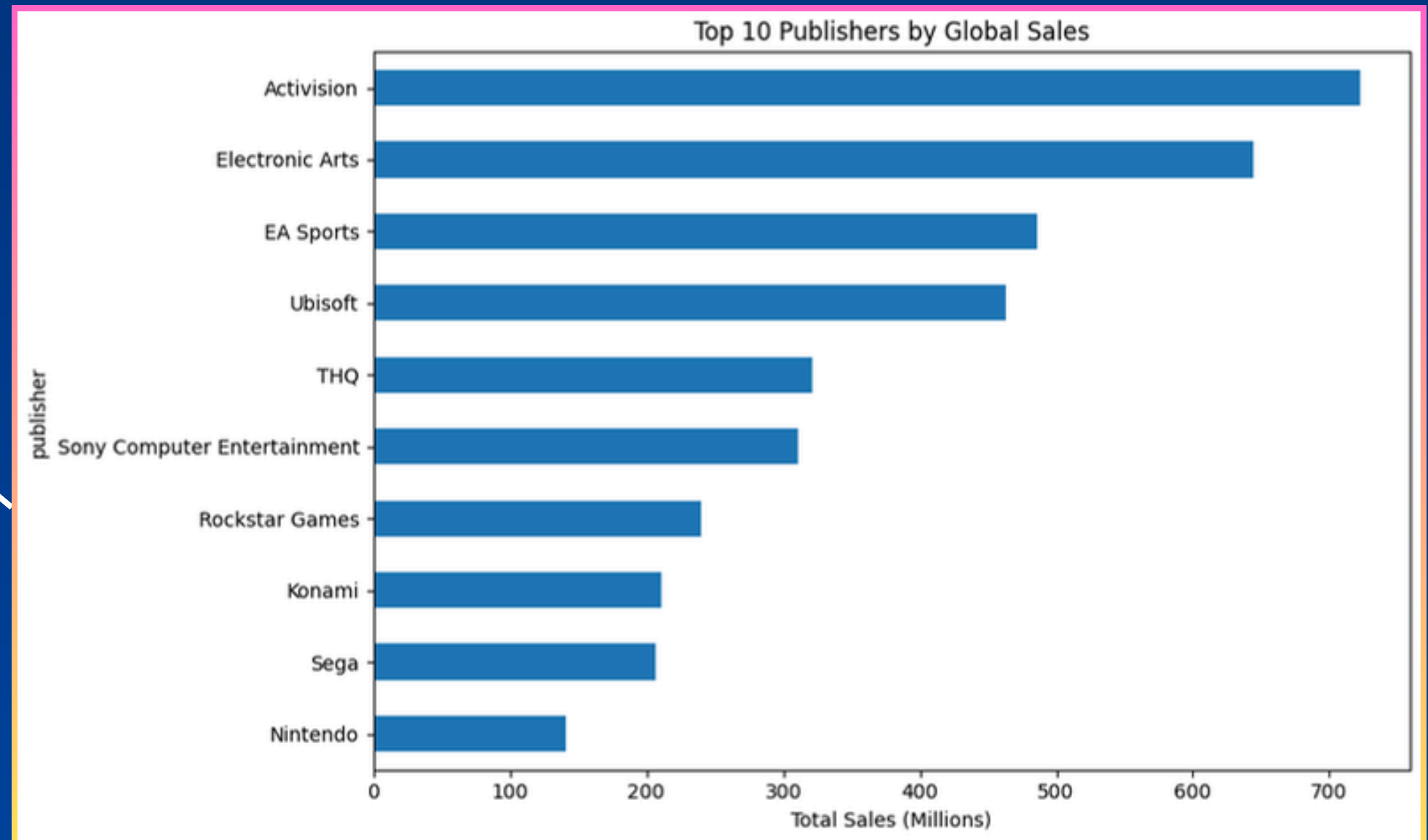


TOP PUBLISHERS VS TOTAL SALES

To determine the largest publisher by the amount of sales, we generated a horizontal bar chart.

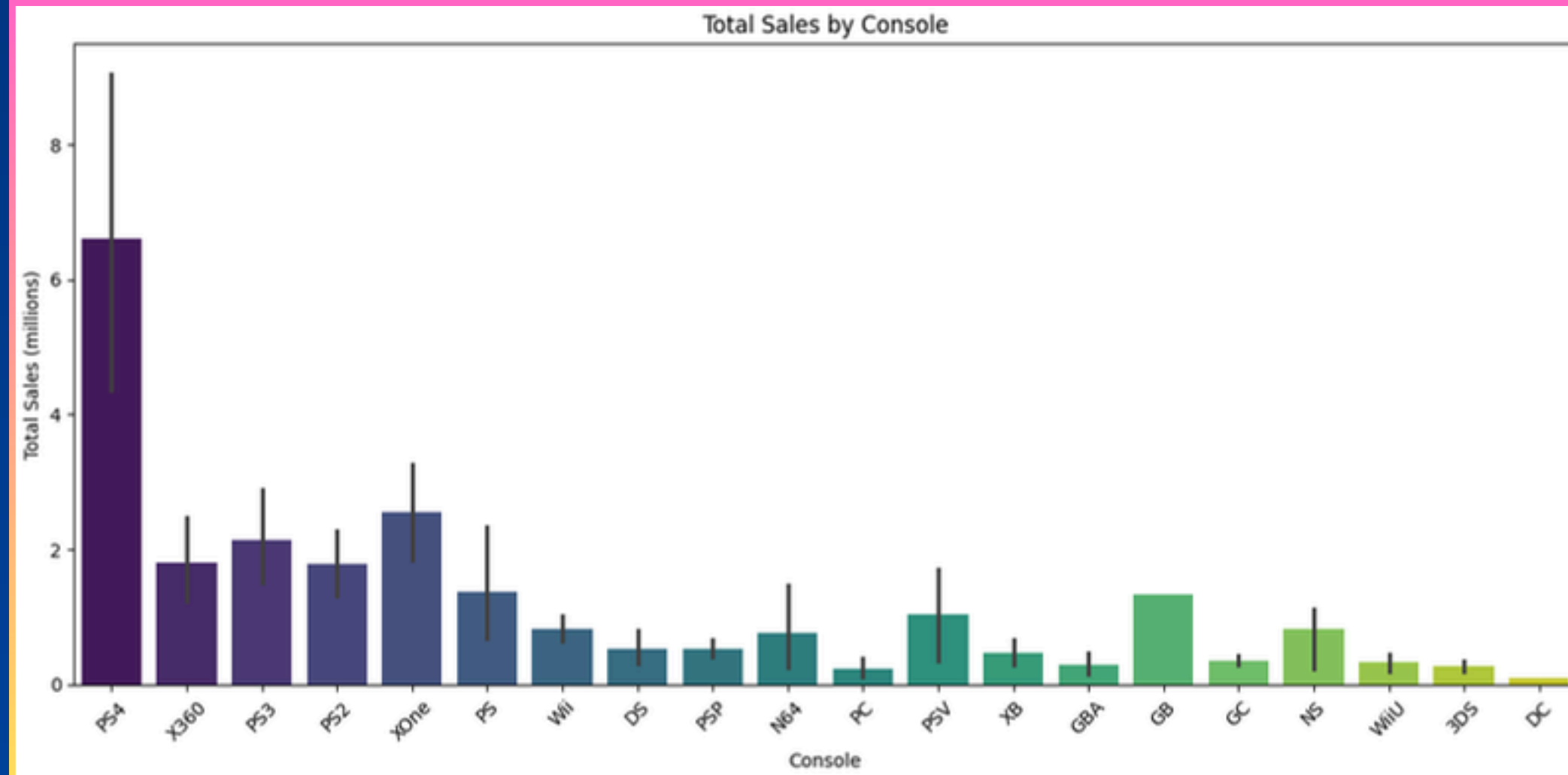
According to the data, **Activision**, known mostly for their **Call of Duty Franchise**, has dominated the market with a total of about **700 million sales**, followed by EA with 650 million sales and EA Sports with **500 million sales**. Smaller publishers appear far below, showing the market concentration among the few giants.

```
top_publishers = df.groupby('publisher')['total_sales'].sum().sort_values(ascending=False).head(10)
top_publishers.plot(kind='barh', title='Top 10 Publishers by Global Sales', figsize=(10, 6))
plt.xlabel('Total Sales (Millions)')
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```



CONSOLE AND TOTAL SALES

```
plt.figure(figsize=(12, 6))
sns.barplot(data=filtered_df, x="console", y="total_sales", palette="viridis")
plt.title("Total Sales by Console")
plt.xlabel("Console")
plt.ylabel("Total Sales (millions)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

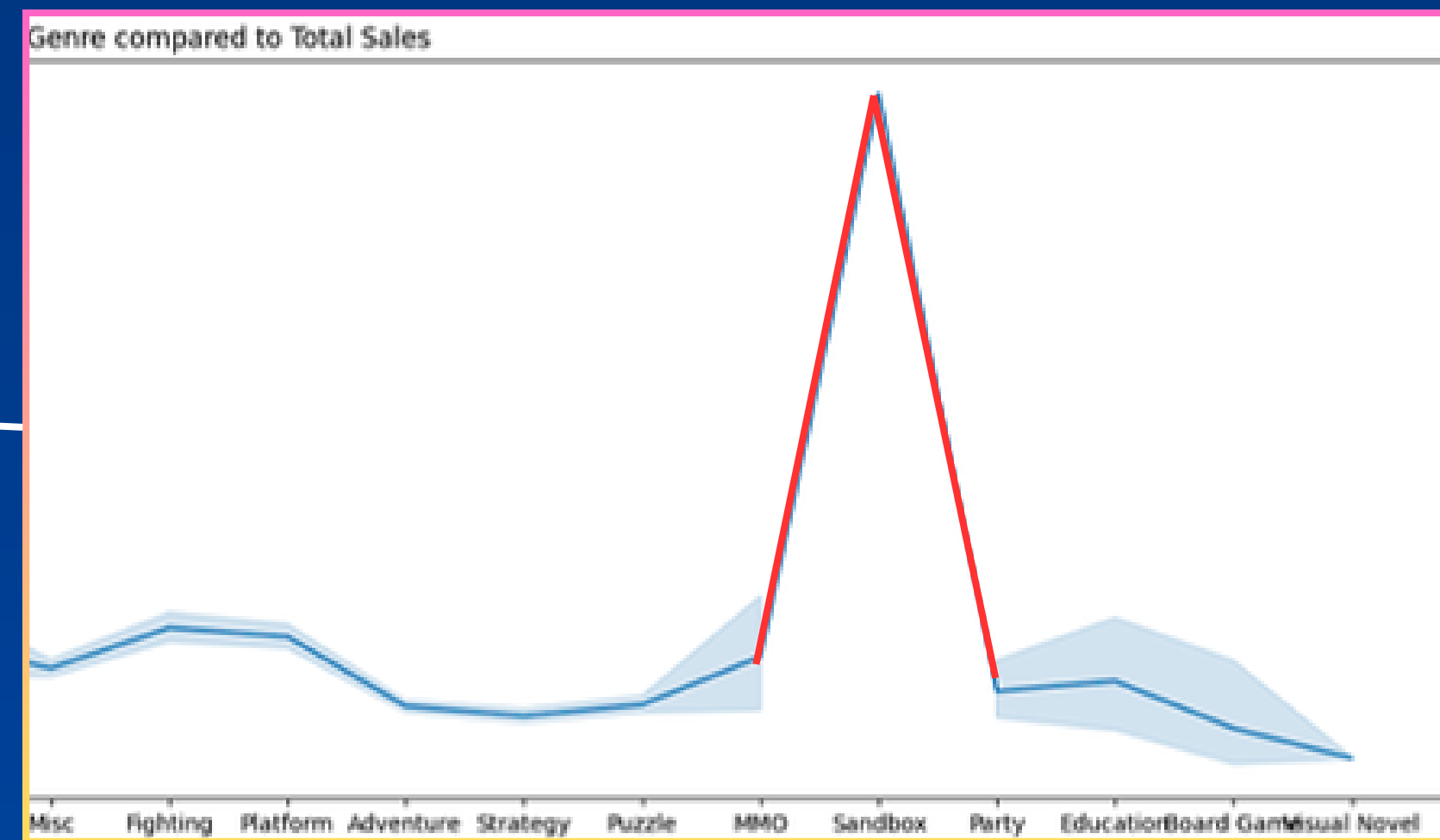
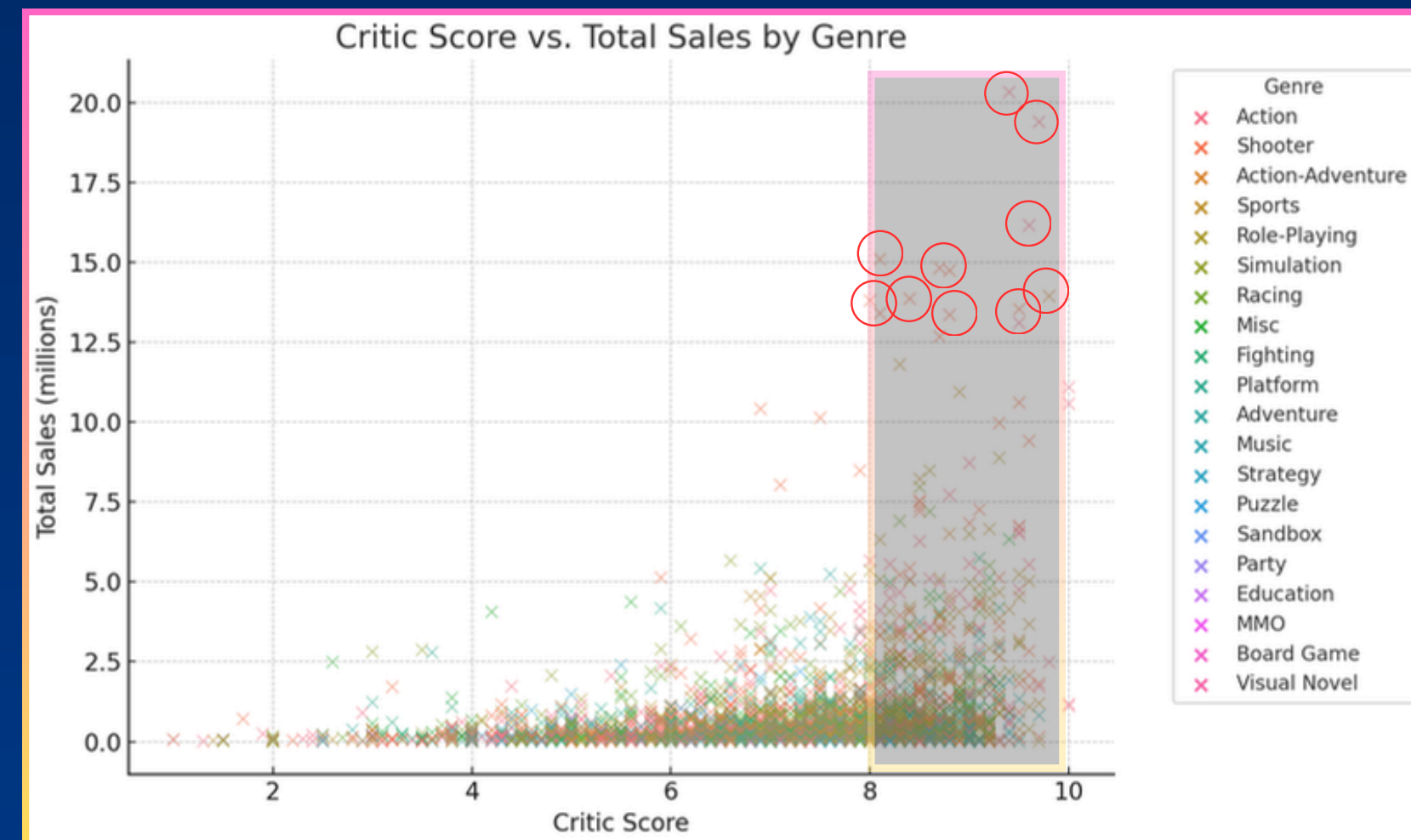


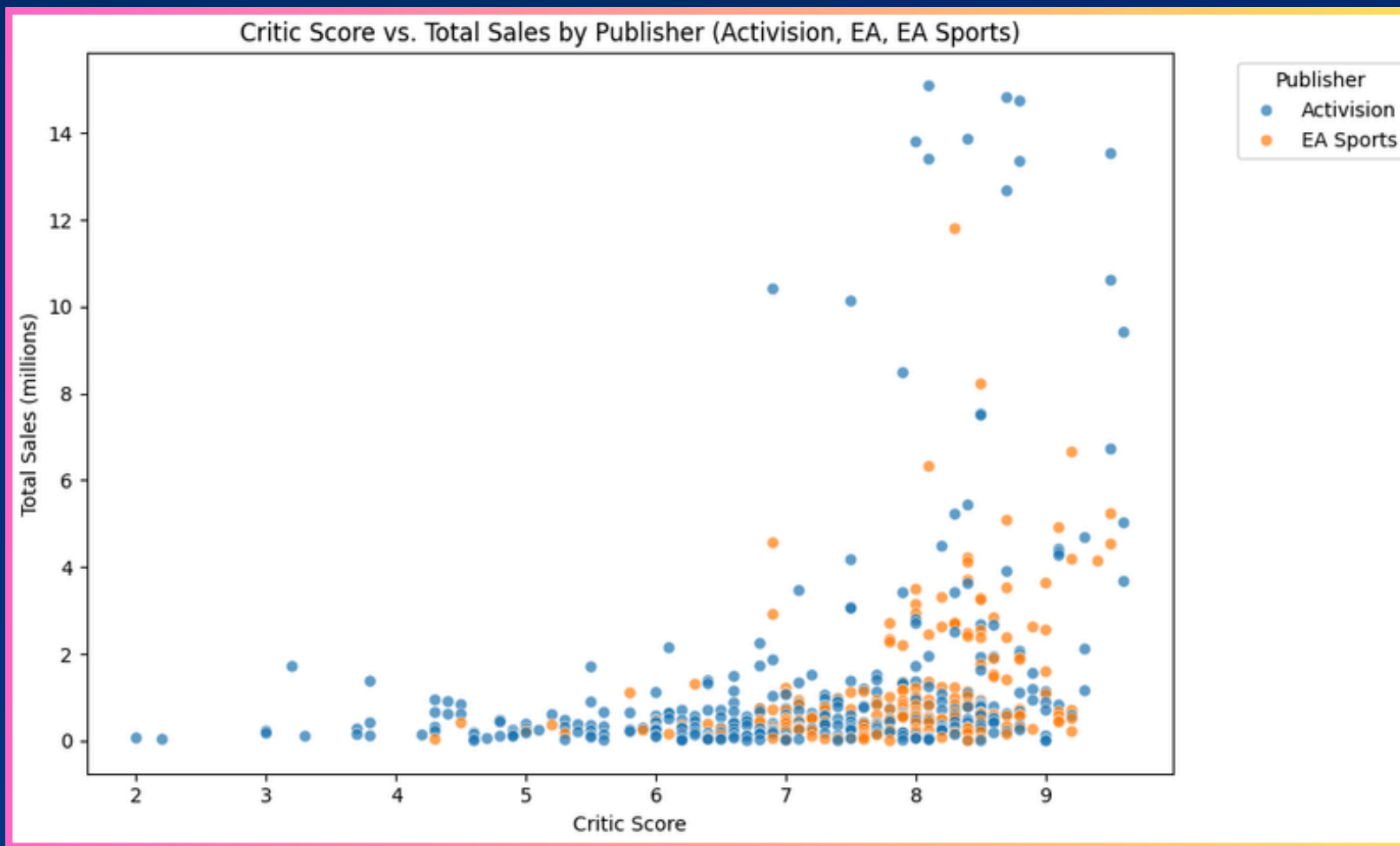
To determine if the platform had any correlation with the sales, we visualized the data using a bar graph

As you can see on the bar graph, games that were released on the PS4 surpassed every single console by a considerable amount, suggesting that a specific platform may be more favorable when it comes to a games success

Critic scores and total sales by genre show how game quality affects success. If high-scoring games in certain genres consistently have strong sales, it suggests that good reviews contribute to a game's commercial performance.

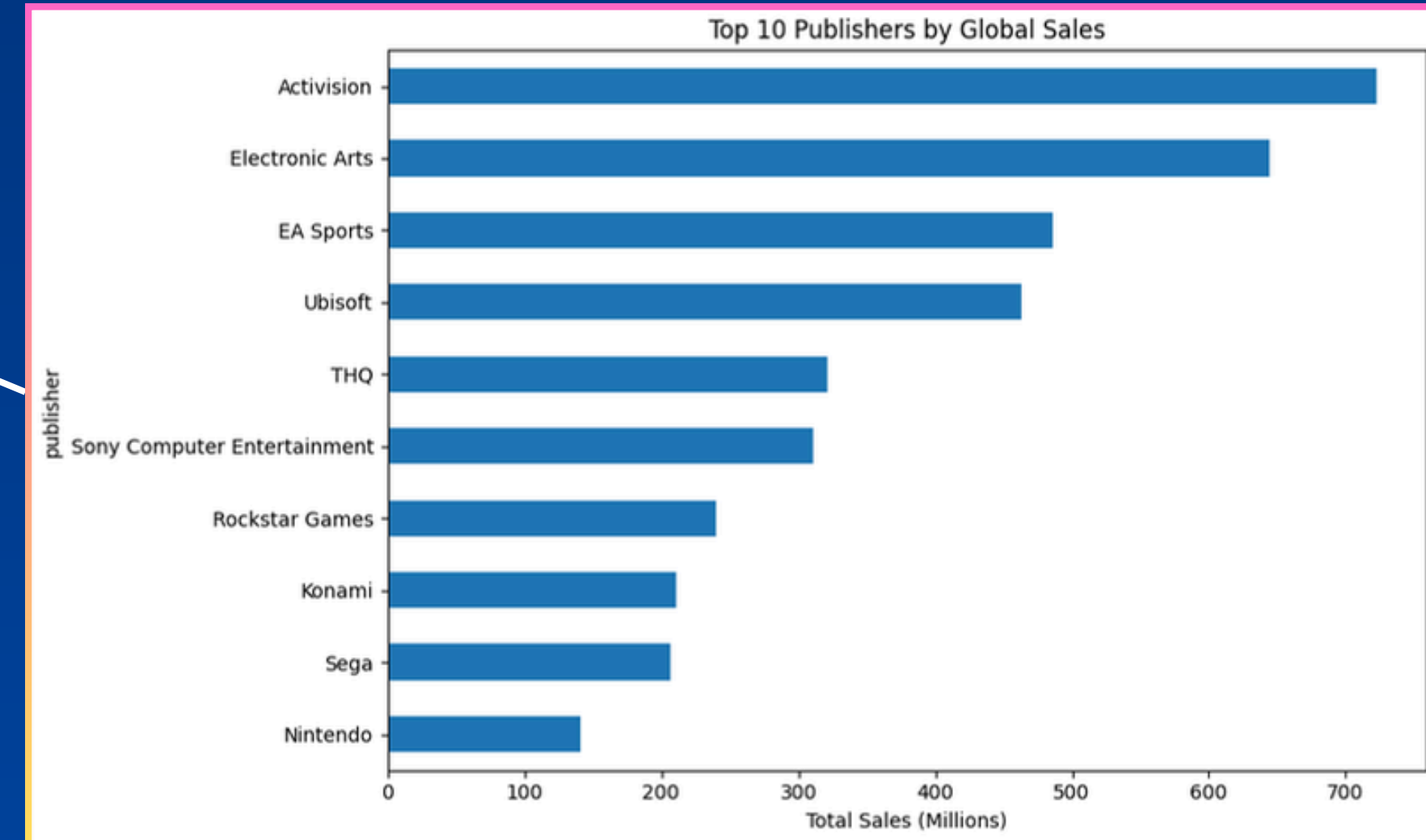
Despite having a perceivably higher total sale, Sandbox games are rarely critically acclaimed, thus suggesting that total sales may not correlate to the positive feedback from the public, which can hinder success, so genre is not a key factor





With the previous deduction, we can see that **major publishers** often have the **higher scoring** and **higher grossing** games

These **Publishers** have **dominated** the market, so **reputation** is definitely **instrumental** in a games **success**



The best visualization for this analysis would be the comparison between Critic Score and Total Sales, as it is a linear least-squares regression line, offering a more direct conclusion to be drawn, as opposed to using confounding variables, like the Publisher or the Year

WHAT WE LOOKED FOR

OUR INITIAL HYPOTHESIS



HIGH RATINGS



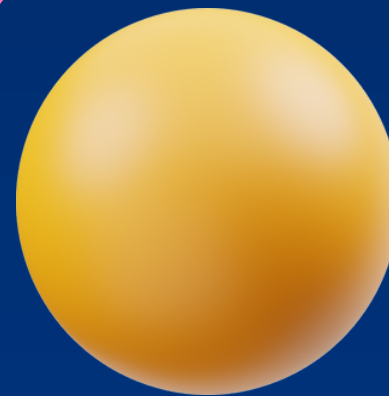
POPULAR GENRE



HIGH SALES

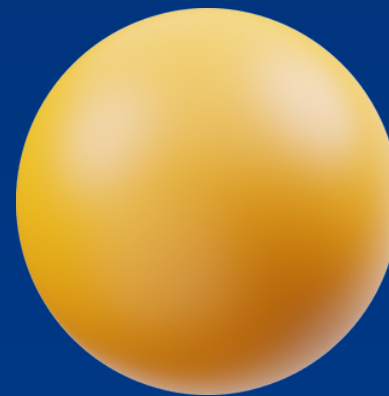
TO ANSWER THE QUESTION?

What ACTUALLY makes a successful game?



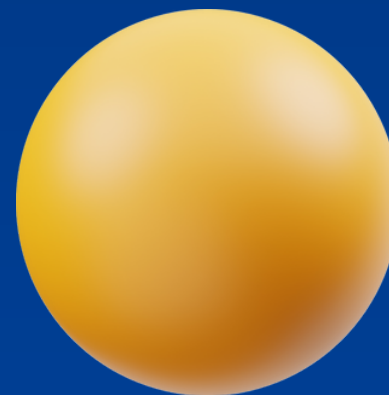
1. High Critic Score

High critic scores often boost visibility, trust, and ultimately sales.



2. High Total Sales

High total sales boost revenue, brand visibility, and enable sequels.



3. Popular Publisher

Popular publishers boost trust, attracts players, and ensures ultimate success



THANK YOU FOR YOUR ATTENTION

Any questions for us?

"Video games are bad for you?
That's what they said about rock
and roll." – Shigeru Miyamoto

Jayden, Ernest, Anthony, Ishita

